

//Selection sort implementation

```
#include <iostream>
using namespace std;

void SelectionSort(int numbers[], int numbersSize) {
    int i, j, indexSmallest, temp;
    for (i = 0; i < numbersSize - 1; ++i) {
        indexSmallest = i;
        for (j = i + 1; j < numbersSize; ++j) {
            if (numbers[j] < numbers[indexSmallest]) {
                indexSmallest = j;
            }
        }
        temp = numbers[i];
        numbers[i] = numbers[indexSmallest];
        numbers[indexSmallest] = temp;
    }
}
```

```
int main() {
    int numbers[] = {10, 2, 78, 4, 45, 32, 7, 11};
    const int NUMBERS_SIZE = 8;

    cout << "UNSORTED: ";
    for (int i = 0; i < NUMBERS_SIZE; ++i) {
        cout << numbers[i] << " ";
    }
    cout << endl;

    SelectionSort(numbers, NUMBERS_SIZE);

    cout << "SORTED: ";
    for (int i = 0; i < NUMBERS_SIZE; ++i) {
        cout << numbers[i] << " ";
    }
    cout << endl;

    return 0;
}
```

```
UNSORTED: 10 2 78 4 45 32 7 11
SORTED: 2 4 7 10 11 32 45 78
```

//Insertion sort implementation

```
#include <iostream>
using namespace std;

void InsertionSort(int numbers[], int numbersSize) {
    int i, j, temp;
    for (i = 1; i < numbersSize; ++i) {
        j = i;
        while (j > 0 && numbers[j] < numbers[j - 1]) {
            temp = numbers[j];
            numbers[j] = numbers[j - 1];
            numbers[j - 1] = temp;
            --j;
        }
    }
}

int main() {
    int numbers[] = {10, 2, 78, 4, 45, 32, 7, 11};
    const int NUMBERS_SIZE = 8;

    cout << "UNSORTED: ";
    for (int i = 0; i < NUMBERS_SIZE; ++i) {
        cout << numbers[i] << " ";
    }
    cout << endl;

    InsertionSort(numbers, NUMBERS_SIZE);

    cout << "SORTED: ";
    for (int i = 0; i < NUMBERS_SIZE; ++i) {
        cout << numbers[i] << " ";
    }
    cout << endl;

    return 0;
}
```

```
UNSORTED: 10 2 78 4 45 32 7 11
SORTED: 2 4 7 10 11 32 45 78
```

//Bubble sort implementation

```

#include <iostream>
using namespace std;

void BubbleSort(int A[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (A[j] > A[j + 1]) {
                int temp = A[j];
                A[j] = A[j + 1];
                A[j + 1] = temp;
            }
        }
    }
}

```

```

int main() {
    int A[] = {2, 7, 4, 1, 5, 3};
    BubbleSort(A, 6);
    for (int i = 0; i < 6; i++)
        cout << A[i] << " ";
    cout << endl;

    return 0;
}

```



1 2 3 4 5 7